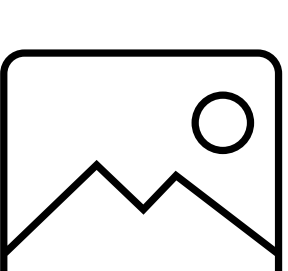


SmartLeafDetection

Based Hybrid Deep Learning Model for Tomato Leaf Disease Detection

Noy Ben Ezra & Ariel Laniado | Advisor: Dr. Dan Lemberg & Dr. Elena Kremer

Motivation

-  Early detection prevents disease spread.
-  Manual inspection is slow and inefficient.
-  Variable conditions hinder reliable leaf detection.

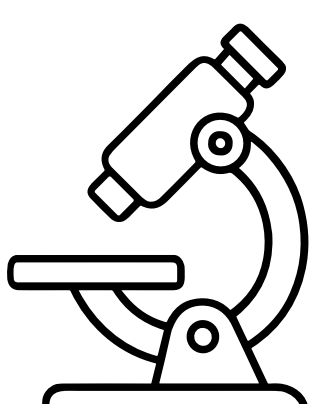
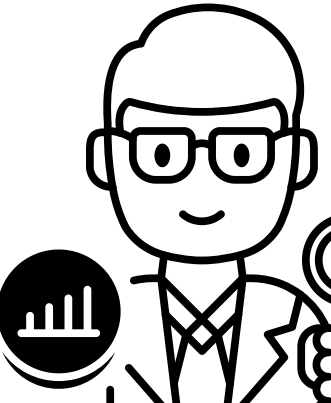
Our platform transforms UAV footage into accurate plant disease detection.

Our Solution

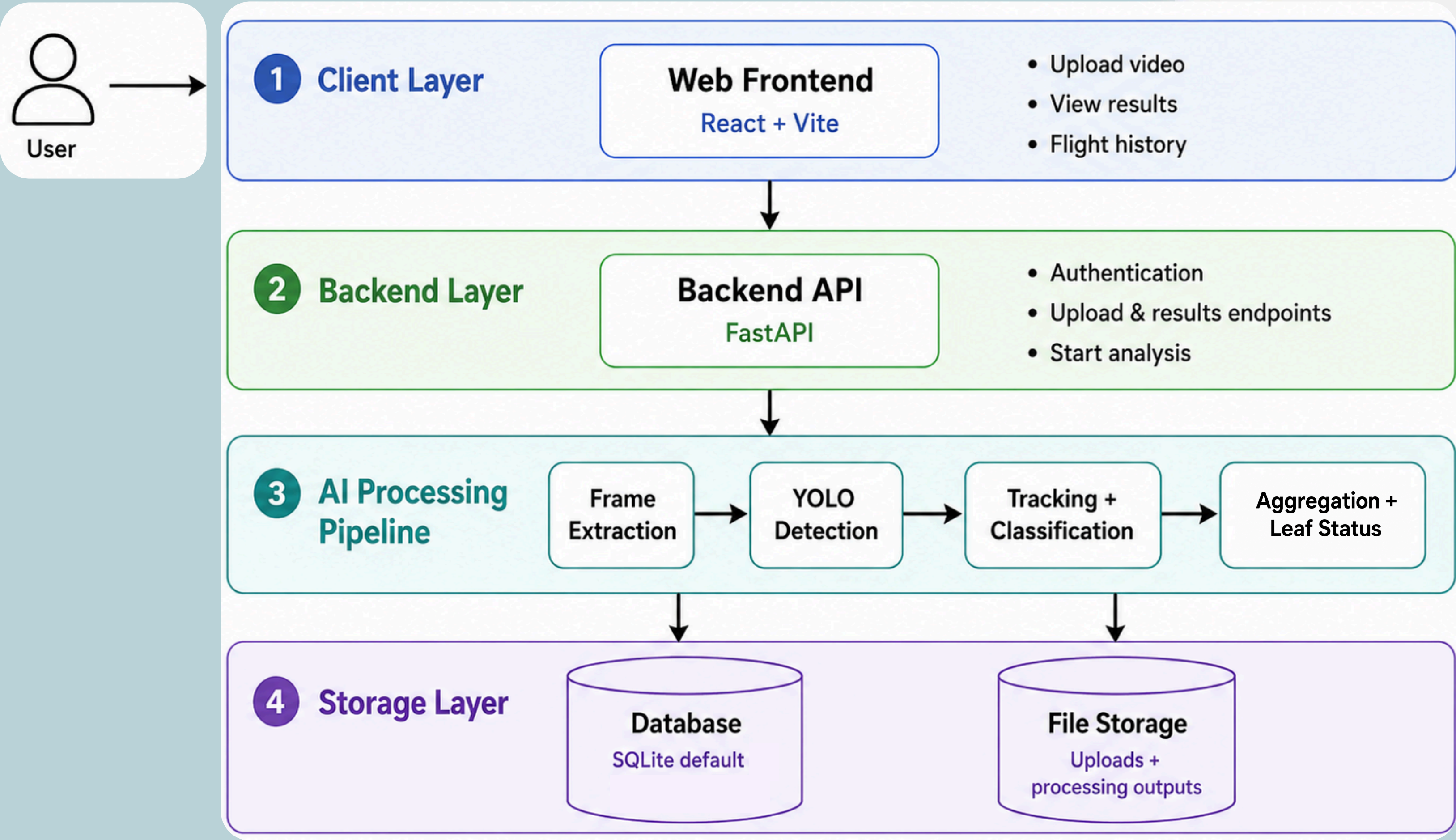
- Two-stage YOLOv11 detection
- Bot-Sort-based tracking
- ResNet50 disease classification
- Temporal aggregation
- Web-based result management

Provides reliable and consistent plant disease diagnosis from UAV videos.

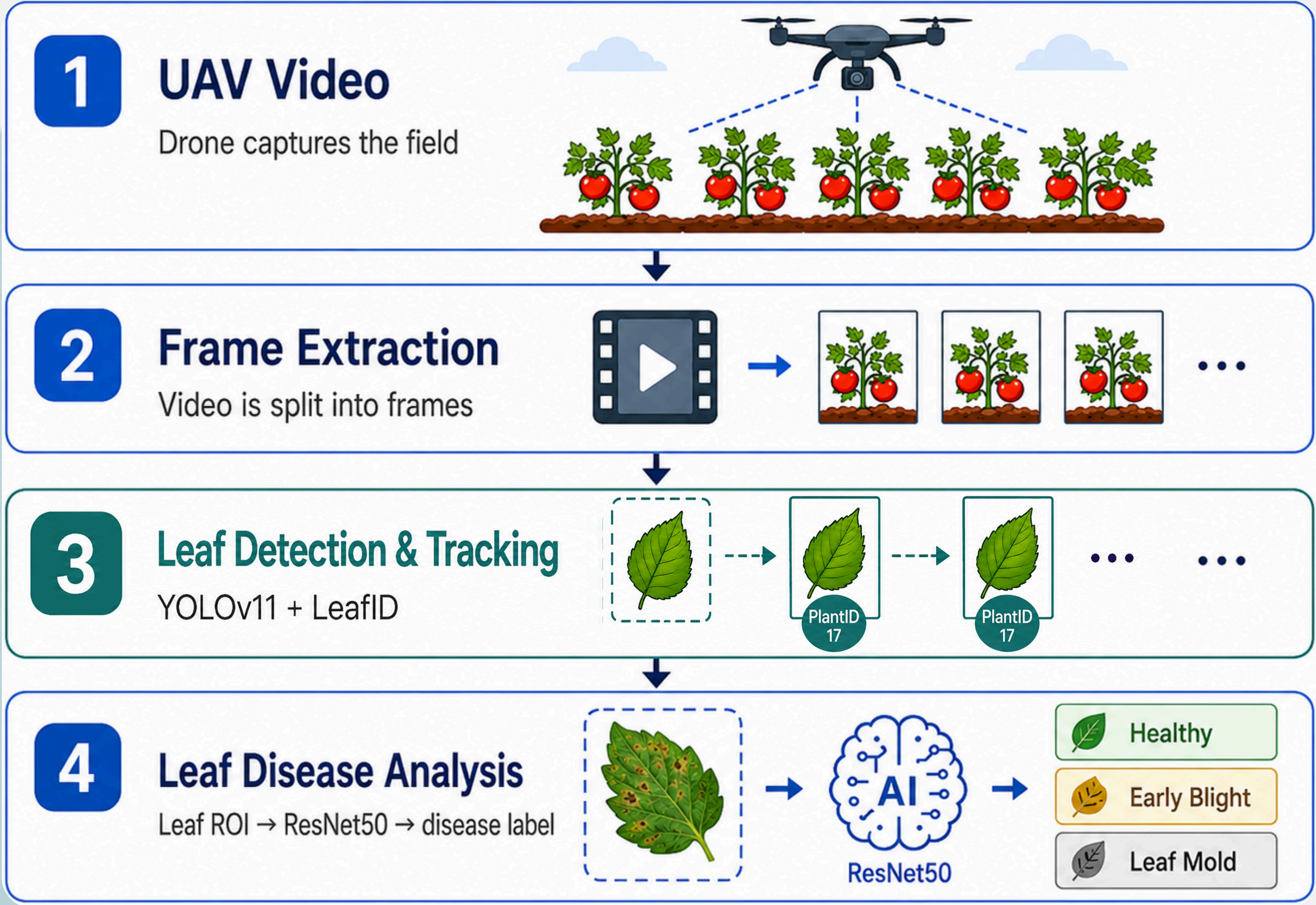
Target Users

-  Farmers
Early disease detection and targeted crop treatment.
-  Agronomists
Field monitoring and disease analysis.
-  Researchers
Field disease detection evaluation.

System Architecture



System Workflow



Expert Insight: Yaakov Rozenberg

Veteran Farmer and Leading Expert in Tomato Cultivation

“What is detected early can be saved in time. Once the disease is visible to everyone, the cost is already high.”

Results & Conclusion

The system integrates detection, tracking, classification, and temporal aggregation into an end-to-end pipeline

Component	Metric	Result
Leaf Detection- YOLOv11s	Recall	81.8%
Disease Classification- ResNet50	Accuracy	94.2%
Disease Classification- ResNet50	Macro-F1	84.9%

Video-based leaf analysis enables reliable and practical disease monitoring

Main Application Screens

